

Net models for wirelength

Multi-pin nets need a model before you optimize

The idea

- Bbox is cheap and standard
- Clique overestimates affinity on multi-pin nets
- Star depends on the hub choice
- Use one model for the reported objective and say which

Pseudocode

- Net models change which edges enter the sum
- Pseudocode names bbox, clique, and star so reports stay comparable
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On golden ABCD, clique HPWL is sixteen and star from A is eight
- Mixing models in one table is a metric bug

Algorithm sketch — try these

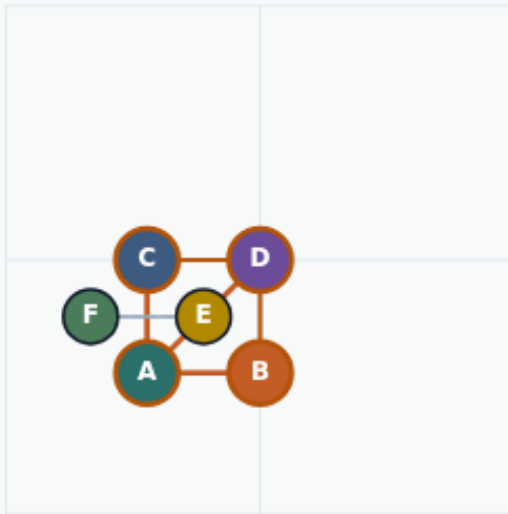
```
INPUT: net pins, positions, model  
bbox: one HPWL on all pins  
clique:  $\Sigma$  pairwise HPWL over pin pairs  
star:  $\Sigma$  HPWL(hub, other pins)  
GOLDEN 4-pin ABCD on golden place:  
    clique=16; star from A=8
```

Bbox HPWL is the default report

NET MODELS

Bbox HPWL is the default report

Step 1 / 5 · bbox-default · module01-03-net-models



Explanation

On the golden four-pin net A-B-C-D, the bounding box is two by two, so bbox HPWL is four. That is the cheap, standard teaching metric.

- Bbox = half-perimeter of pin hull
- Cheap to compute
- Standard report across labs

ABCD bbox HPWL: 4
Total golden HPWL: 14

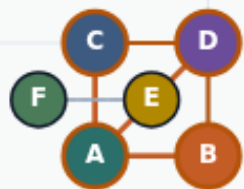
Bbox HPWL is the default report

Clique sums every pair

NET MODELS

Clique sums every pair

Step 2 / 5 · clique-model · module01-03-net-models



Explanation

Clique wirelength adds HPWL of every pairwise edge among the four pins. On golden ABCD that sum is sixteen —four times the bbox number.

- $C(n,2)$ pairwise spans
- Overestimates multi-pin affinity
- Same pins, larger objective

```
clique(ABCD): 16  
bbox(ABCD): 4  
Ratio: 4× on this instance
```

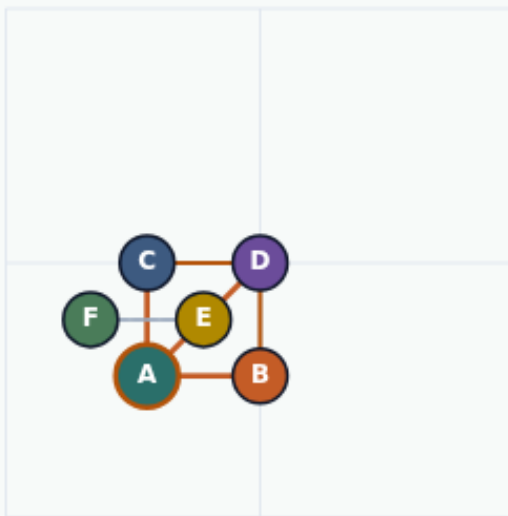
Clique sums every pair

Star depends on the hub

NET MODELS

Star depends on the hub

Step 3 / 5 · star-model · module01-03-net-models



Explanation

Star sums spokes from a hub to every other pin. With hub A on golden ABCD, star HPWL is eight—between bbox four and clique sixteen.

- Pick a hub (here A)
- Sum HPWL(hub, other) for each pin
- Hub choice changes the number

```
star-from-A: 8  
clique: 16  
bbox: 4
```

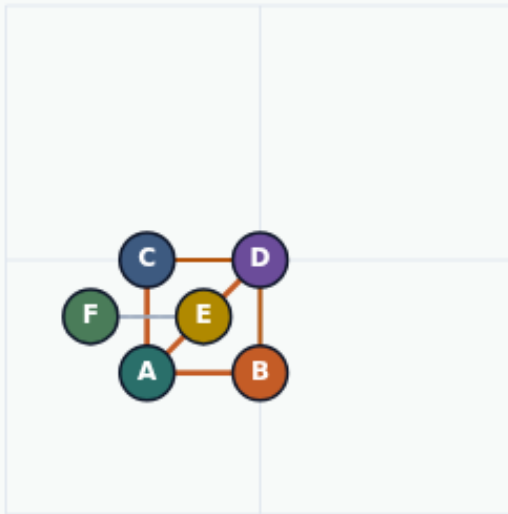
Star depends on the hub

Same pins, three numbers

NET MODELS

Same pins, three numbers

Step 4 / 5 · compare-models · module01-03-net-models



Explanation

Bbox four, star-from-A eight, clique sixteen. Pick one model for the reported objective and say which—mixing models is how goldens quietly disagree.

- Bbox: cheap standard
- Star: hub-sensitive
- Clique: densest affinity

```
bbox = 4  
star-from-A = 8  
clique = 16
```

Same pins, three numbers

Name the model you optimize

NET MODELS

Name the model you optimize

Step 5 / 5 · takeaway · module01-03-net-models



Explanation

Use one wirelength model in the objective and in the report. The rest of the course quotes bbox HPWL unless a lab explicitly asks for clique or star.

- Default report = bbox HPWL
- Golden ABCD: 4 / 8 / 16
- Next: force-directed place

```
cliqueHpwlGolden4pin: 16  
starHpwlGolden4pinFromA: 8
```

Name the model you optimize

Browser lab track

- In the browser lab track, open the **net-models** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse ``tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module